Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I		**Measures of dispersion**	**10**	**10**
	1	Range	1	
	2	Quartile deviation	3	
	3	Standard deviation	3	
	4	Coefficient of variation	3	
		Unit 1: 2.1.3 Ref[1] Unit 2: 2.1.3 Ref[1] Unit 3: 2.1.3 Ref[1] Unit 4: 2.1.3 Ref[1]		
II		**Correlation and regression**	**13**	**20**
	5	Bivariate distribution, correlation	1	
	6	Scatter diagram	2	
	7	Karl Pearson coefficient of correlation	2	
	8	Limits of Correlation coefficient	2	
	9	Regression	2	
	10	Lines of regression	3	
	11	Regression coefficients	2	
		Unit 5: 10.1 Ref[2] Unit 6: 10.1Ref[2] Unit 7: 10.3 Ref[2] Unit 8:10.3.1 Ref[2] Unit 9: 10.7 Ref[2] Unit 10: 10.7.1 Ref[2] Unit 11:10.7.3 Ref[2]		
III		**Probability theory**	**10**	**25**
	12	Random experiment	1	
	13	Samplespace	1	
	14	Event	1	
	15	Classical Probability-definition	2	
	16	Statistical probability-definition	2	
	17	Axiomatic approach to Probability	2	
	18	Addition theorem (Statement only)	1	
		Unit 12:4.5.1 Ref[2] Unit 13: 4.5.1 Ref[2] Unit 14: 4.5.2 Ref[2] Unit 15: 4.3.1 Ref[2] Unit 16: 4.3.2 Ref[2]		

	Unit 17: 4.5 Ref[2] Unit 18: 4.6.2 Ref[2]			
IV		**Conditional Probability**	**12**	**15**
	18	Conditional Probability of two events	3	
	19	Multiplication theorem (Statement only)	2	
	20	Independence of events	2	
	21	Conditions of mutual independence of three events	2	
	22	Bayes theorem and its applications (Statement only)	3	
	Unit 18: 4.7 Ref[2] Unit 19: 4.7 Ref[2] Unit 20: 4.7.3 Ref[2] Unit 21: 4.7.5 Ref[2] Unit 22: 4.8 Ref[2]			